Live on Cloud Run · Google Cloud AI Builder Cup 2026

Naqsha نقشہ

Walk through your home before the first brick.

From a rough idea to an LDA-checked floor plan and a 3D home you can walk through, comment on and change.

Gemini 4 Argon designs the house in code. A Google ADK agent revises it. All on Google Cloud.



naqsha-242926648187.us-central1.run.app

github.com/abdm693/hack2skill-hackathon

[Team name] · Lahore first · LDA rules of 9 Feb 2026

In Lahore, a house starts as a sketch and a guess

Johar Town · 10 marla (35′ × 65′, about 2,275 sq ft)

A family of six wants four bedrooms on one budget. They start with a paper sketch and a contractor's guess, then weeks of back-and-forth with an architect.

Nobody walks through it until it's built.

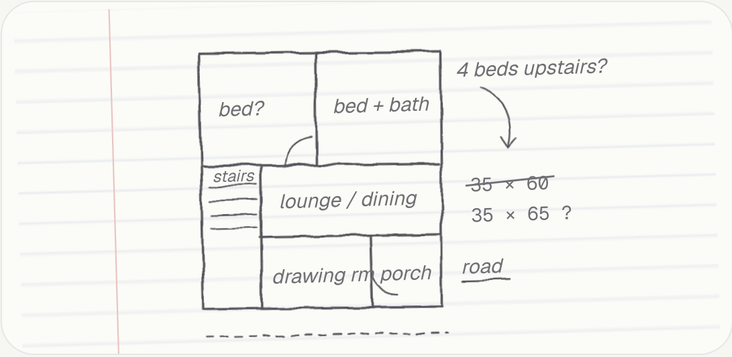
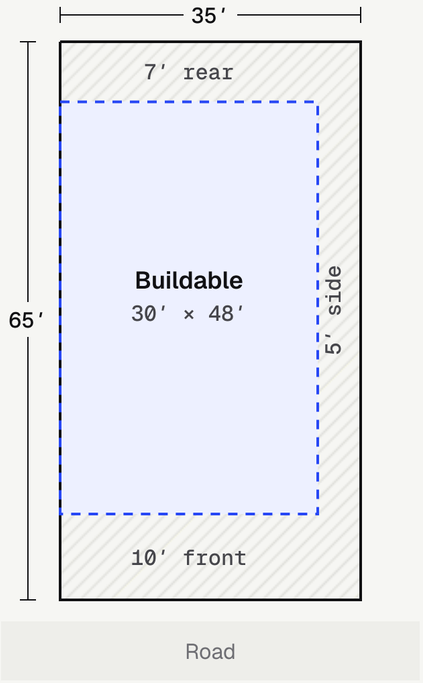


Illustration: how many plans begin, on paper.



Same plot, by the rules.

What that plot must respect under the Lahore Development Authority's (LDA) rules of 9 Feb 2026

10′ · 7′ · 5′	open at the front, the rear and one side
70%	ground coverage, at most
45′	to the top of the parapet
3 storeys	down from 4 in the 2020 text
PCATP stamp	an architect of the Pakistan Council of Architects and Town Planners signs it
Rs 15 / sq ft	scrutiny fee, paid again on every revised plan

What a mistake costs on this plot · worked out from LDA's fee schedule

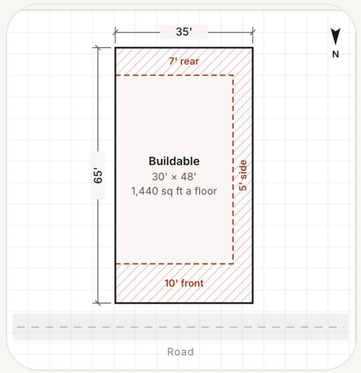
≈ Rs 38,000	for each resubmission of the 2,565 sq ft house Naqsha designed for this plot
Rs 15–30k	penalty for a porch 4 ft into the front open space: 60 sq ft at Rs 250 or Rs 500

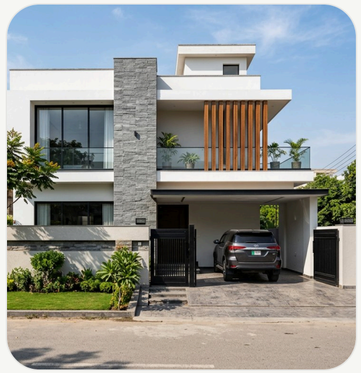
The rules are written down. They just aren't in the room when the first sketch is drawn.

Source: LDA Building & Zoning Regulations 2019, consolidated text of 9 Feb 2026, Appendix C

One sitting, from an idea to a house you can walk through

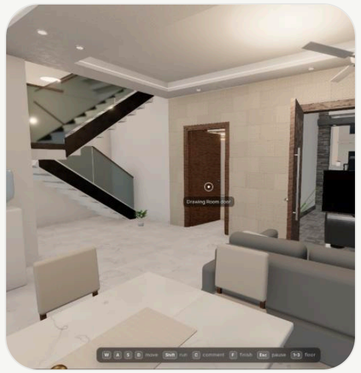
Brief, concepts, plan, 3D, walk and revise. Shown on the live 10-marla sample.











Car porch, ground floor

“Can we extend the car porch 4 feet forward into the front lawn?”

● Not possible

“…a mandatory 10 ft front setback is required and the car porch slab cannot project into the front open space.”

~7 min, plan + LDA checks + 3D

v2, v3 …

Brief

instant

Answer one-tap questions, or skip and let the AI choose. The LDA envelope appears as soon as the plot is known.

`gemini-3.8-flash + deterministic envelope`

Concepts

12–17 s

Four photoreal exteriors, generated in parallel. Pick one, or remix it in plain words.

`gemini-3.1-flash-image × 4`

Plan

A dimensioned plan, checked against LDA rules and fixed by Argon until it passes.

`gemini-4-argon + LDA engine`

3D

Argon writes the house as code, and it assembles live: structure, facade, interiors, furniture, landscape.

`gemini-4-argon → House SDK code`

Walk

live

First person, with collisions and stairs. Pin a comment on any surface.

`three.js + three-mesh-bvh`

Revise

~95 s

A new version with what changed, or the rule that says no and an alternative.

`Google ADK agent, 9 tools`

Times measured on the live Cloud Run service · sample homes open instantly, with no wait

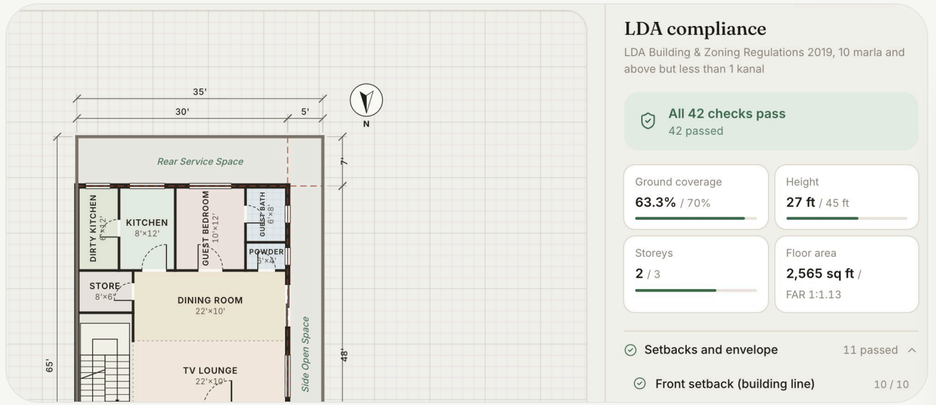
It's live on Cloud Run today

Open a sample home and walk it in a minute.

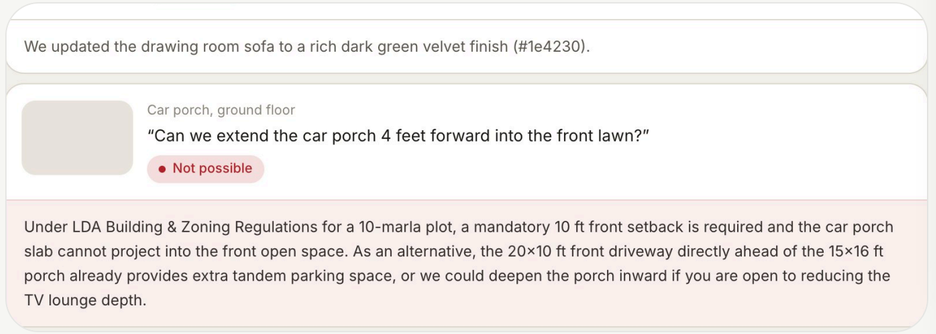
naqsha-242926648187
.us-central1.run.app



3D. The 1-kanal sample, a second brief: Argon wrote it as 25 code sections.



Plan. The 10-marla sample: all 42 LDA checks pass.



Revise. A real ADK run: the porch refused, with the rule and an alternative.



Phone. The same 3D house.

Captured from the live service on 17 Oct 2026, from the two public sample homes

Argon designs in code, not pixels

So the house is exact, checkable, editable one section at a time, and walkable.

Brief
+ the chosen concept

1 Plan — gemini-4-argon

A schema-bound JSON plan, in feet: levels, rooms, doors, windows, stairs, roof, mumty and boundary wall.

check ↓ / fix, ≤ 3 attempts ↑

LDA engine — deterministic

18 check types, run per room and stair, from LDA's text of 9 Feb 2026

2 Build — design.js · House SDK

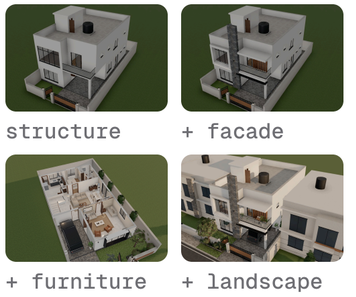
Inputs: plan JSON · concept image · SDK types

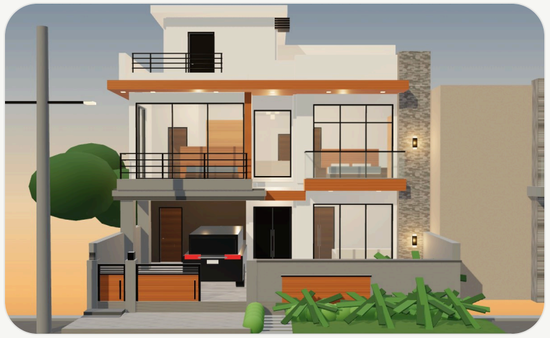
Argon writes 23–25 named sections against our SDK: 44 PBR presets, 57 furniture types, real wall openings, and raw three.js where a detail is bespoke.

structure · facade_stone_pier · facade_left_bay · interior_ground · furnish_g_lounge · landscape · lighting · … each tagged {layer, level, room}

3 Stream

renders after every section





Before: one shot, raw three.js. 1,780 lines in one block.



After: House SDK, in sections. The outlined pier is one section, `facade_stone_pier`.

Same model, two ways of working

481 s	for the one shot: raw three.js, nothing to see until the end, and a blocky house
~7 min	today, for the plan, the LDA checks and a high-fidelity house that renders after every section

The whole pipeline now takes about as long as the one shot spent writing blocky code alone.

One-shot spike: 16 Oct 2026, re-shot from a matching front camera · both are 10-marla Lahore houses by Argon

Argon is creative. The rules are code.

A deterministic engine checks every plan, and enforces safety whatever Argon writes.

LDA engine · Live

● 42/42 LDA checks

on the 10-marla sample. Plans run 42–50 checks, depending on their rooms and stairs.

18 check types, applied per room and per stair: setbacks, coverage, FAR, height, storeys, 8% daylight and stairs.

Argon's fix loop, measured live

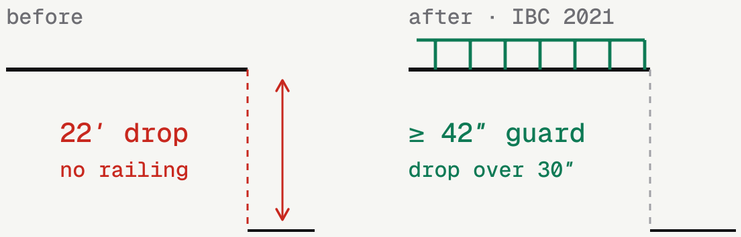
5 → 1 → 0

attempt 1: 5 · attempt 2: 1 · attempt 3: 0

And **4 → 0** on the 1-kanal sample: three daylight failures and one undersized room, fixed in one retry.

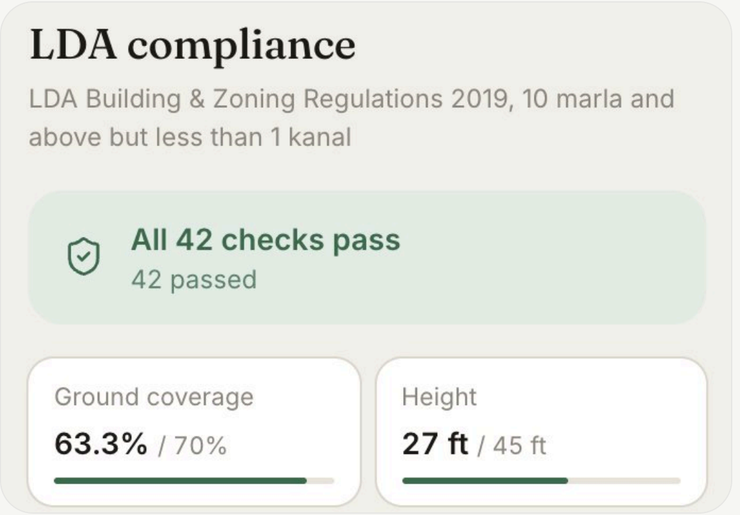
Livable-home rules · ○ In the 18 Oct build

- Bedrooms open only to a corridor, lounge, balcony or their own bath.
- A stair reaches every floor, roof included.
- A railing at every drop, enforced by the engine.
- Doors kept clear of furniture, with a **3′** path.



We audited three houses (two by Argon, one hand-made). All three had stair-safety defects. LDA doesn't define guards, so we apply IBC 2021.

The live compliance panel



Argon self-review · ○ In the 18 Oct build

Argon looks at a rendered image of its own plan and critiques circulation, privacy, daylight and the brief before the family sees it.

Comment in 3D. The agent revises, or tells you which rule says no.

An agent, not one prompt: it reads only what it needs, dry-runs every layout change, and saves nothing until it finishes.



What the agent changed · v1 → v2

plan	f_master_w1 · window 10×7 → 13×8.5 ft
code	furnish_f_master · curtains to 13 ft
code	furnish_g_drawing · green velvet sofa
check	42/42 LDA · still pass

Google ADK 2.12 · gemini-4-argon

Look
get_room · read_section
list_sections

↓

Test
check_plan_ops
a dry run on the LDA engine

↓

Change only what's needed
apply_plan_ops
write_section · remove_section

↓

Explain
resolve_comment · finish

↑ back to Look for the next comment

If the ADK run fails, a single-shot reviser takes over.

One live run on the 10-marla sample — 3 comments · ~95 s

1 · First floor · master bedroom window — Done

“Please make the master bedroom window bigger, I want much more morning light.”

“We enlarged the master bedroom front window from 10×7 ft to 13×8.5 ft with a low 0.5 ft sill to bring in plenty of morning sunlight.”

2 · Ground floor · drawing room sofa — Done

“Change the drawing room sofa to a dark green velvet one.”

“We updated the drawing room sofa to a rich dark green velvet finish (#1e4230).”

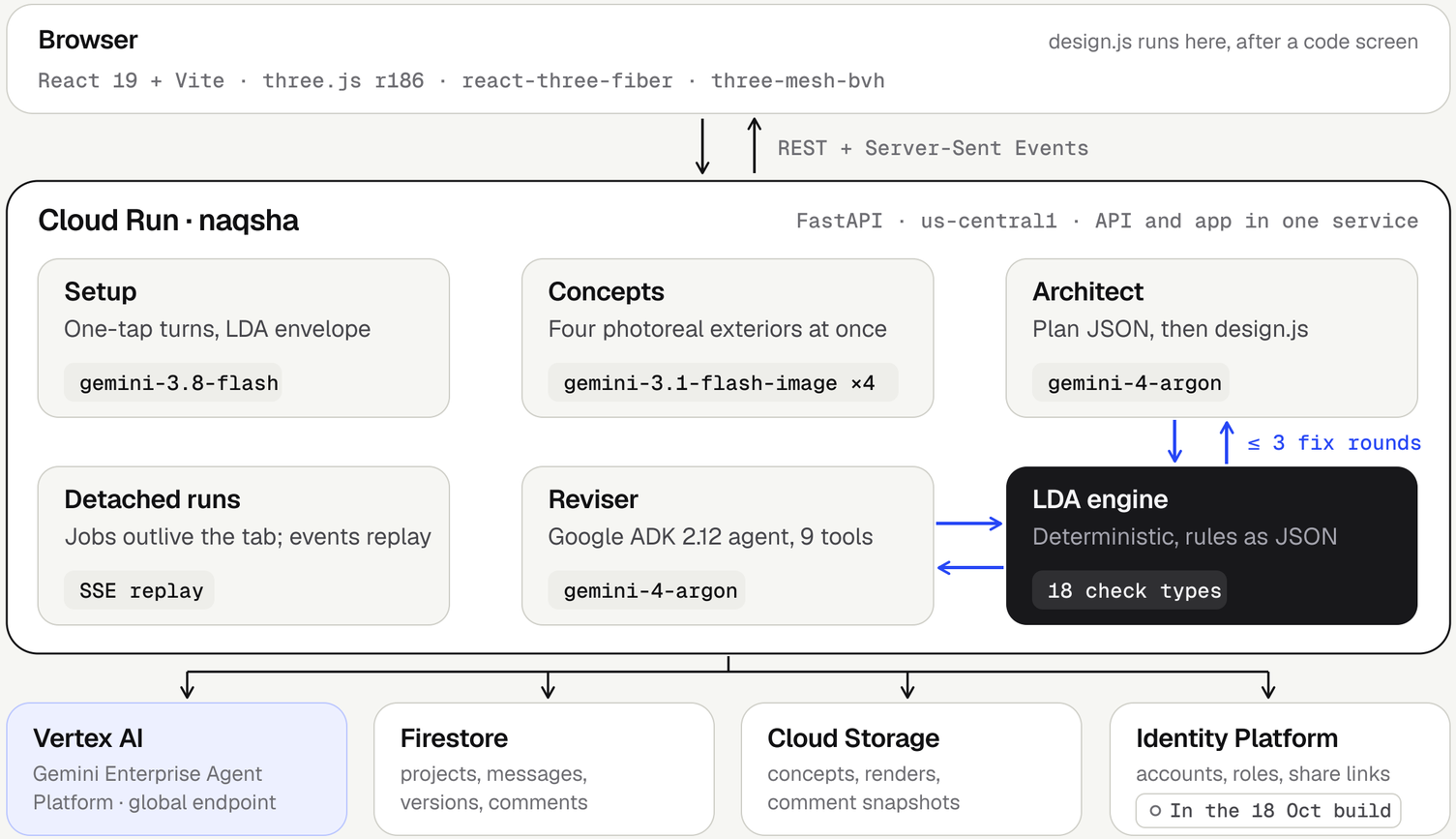
3 · Ground floor · car porch — Not possible

“Can we extend the car porch 4 feet forward into the front lawn?”

“Under LDA Building & Zoning Regulations for a 10-marla plot, a mandatory 10 ft front setback is required and the car porch slab cannot project into the front open space. As an alternative, the 20×10 ft front driveway directly ahead of the 15×16 ft porch already provides extra tandem parking space, or we could deepen the porch inward if you are open to reducing the TV lounge depth.”

Comments and answers verbatim from the live run, 16 Oct 2026 · pins placed at each comment's anchor

One Cloud Run service, Gemini on Vertex AI, state in Firestore



Gemini, matched to each job

Setup `gemini-3.8-flash`
Structured turns and one-tap chips. Fast and cheap.

Concepts `gemini-3.1-flash-image`
Four at once. Speed over hero quality.

Plan `gemini-4-argon`
Schema-bound JSON in feet, reasoned against hard limits.

Build `gemini-4-argon`
Long, correct code against our typed SDK, from the concept.

Revise `gemini-4-argon + ADK`
Nine tools, a dry run before every write, one save at the end.

One service, one URL. No CORS, and a deploy is one `gcloud run deploy --source .`

google-genai 2.31 · google-adk 2.12 · Cloud Run, 3,600 s request timeout

Built for real models on real quotas

Every step of a design run has a guardrail, so a long agent job survives quota errors, bad output and closed tabs.

1 **Gemini call**
Vertex AI · global

2 **Stream**
SSE to the browser

3 **Code screen**
before a section runs

4 **Browser run**
three.js r186

5 **As-built check**
3D model vs. LDA

6 **Revise**
Google ADK 2.12

Backoff on quota errors

429s and 5xx errors retry with exponential backoff and jitter: 6 attempts, from 4 s up to 60 s. google-genai and ADK share the policy.

Degenerate-decoding watch

A runaway tail such as `35.0000000…` aborts and retries the attempt. Prompts cap numbers at 2 decimals.

21 banned patterns

Network, eval, storage, timers and globals are screened out of generated code before it runs.

Self-healing sections

A section that throws is rewritten on its own by Argon. If code is missing, a default design is built from the plan.

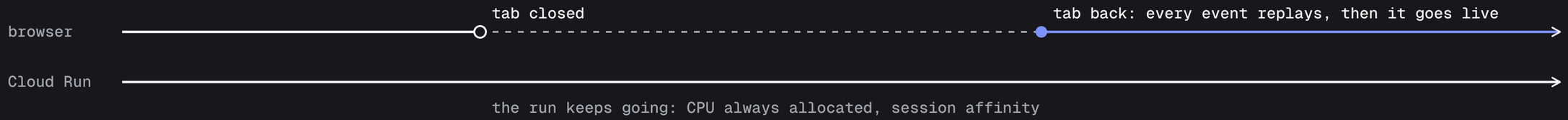
Nothing in a setback

No part of the 3D model may reach into an open space beyond the chajja (sunshade) allowance: 3 ft, or 1.5 ft where the space is under 10 ft.

A fallback reviser

If the ADK run fails, a single-shot reviser takes over. ADK saves nothing until it finishes.

Detached runs A design or revision keeps going when the browser goes away.



Each guardrail is in the public repo: retry policy, stream guard, code screen, repair, as-built check, fallback

Two briefs, two houses, one pipeline, live today

And a short, concrete path from one city to many.



10-marla modern, Johar Town

35′ × 65′ · faces north · 2 storeys · 4 bedrooms

Spaces	28
Code sections	23
LDA checks	42 / 42 pass
Revision	3 comments: 2 done, 1 refused



1-kanal Spanish villa

50′ × 90′ · faces south · 2 storeys · 5 bedrooms

Spaces	34
Code sections	25
LDA checks	50 pass · 2 advisory
Fix loop	4 → 0 in one retry

How it scales

Cloud Run, as deployed

Autoscales to 5 instances today, each 2 vCPU and 2 GiB with 40 concurrent requests, CPU always on. The cap is one flag.

A shared Gemini quota

Every call to the global endpoint backs off on 429s, so a burst queues instead of failing.

Long jobs

Today: an in-memory event log per instance, held by session affinity. Next: the log in Firestore with Cloud Tasks, so any instance can resume.

Cost follows intent

Samples open with no model calls. Briefs and concepts use Flash models; Argon runs only on a full design.

New cities are data

Rules live in one JSON file per authority, and DHA's height and storey caps already switch on.

Evidence: the live samples API and the Cloud Run service settings, read 17 Oct 2026, in the public repo

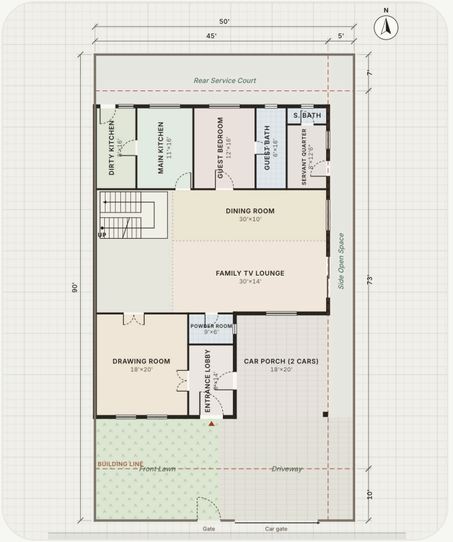
Built for this plot, in this city

The same 1-kanal brief, 50 ft wide. Maket's own FAQ says its designs aren't ready to build; Naqsha checks every plan against LDA rules first.

	Status quo	maket.ai, 16 Oct 2026	Naqsha
Knows the plot	From memory	No: it drew a house wider than the plot	Envelope first; 42–50 LDA checks per plan
Local rooms	Yes	US-centric: no drawing room or servant quarter	Drawing room, dirty kitchen, servant quarter
Floors, stairs	On paper	One floor when free; no stairs anywhere	Up to 3 storeys, checked stairs, a mumty
3D	None until built	White-box massing; a walk through walls	Facade, interiors, furniture as code; collisions
Changes	Redraw by hand	Chat edits at 5 credits each, paid plans	Pin a comment; the agent revises or cites the rule
Speed	Weeks to a drawing	About 6 min to a 2D plan: faster than us to 2D	About 7 min to a walkable, checked 3D house



maket.ai: a 59′6″ house for a 50 ft plot, with no warning.



Naqsha: inside the 10′ front, 7′ rear and 5′ side setbacks.

Source: our maket.ai teardown on its free tier, 16 Oct 2026

The family, the architect and the builder, in one home

One Share dialog. Invite Team or Client by email, or turn on a client link that can comment or only view.

Clients need no account. They give a name, walk the house and pin comments.

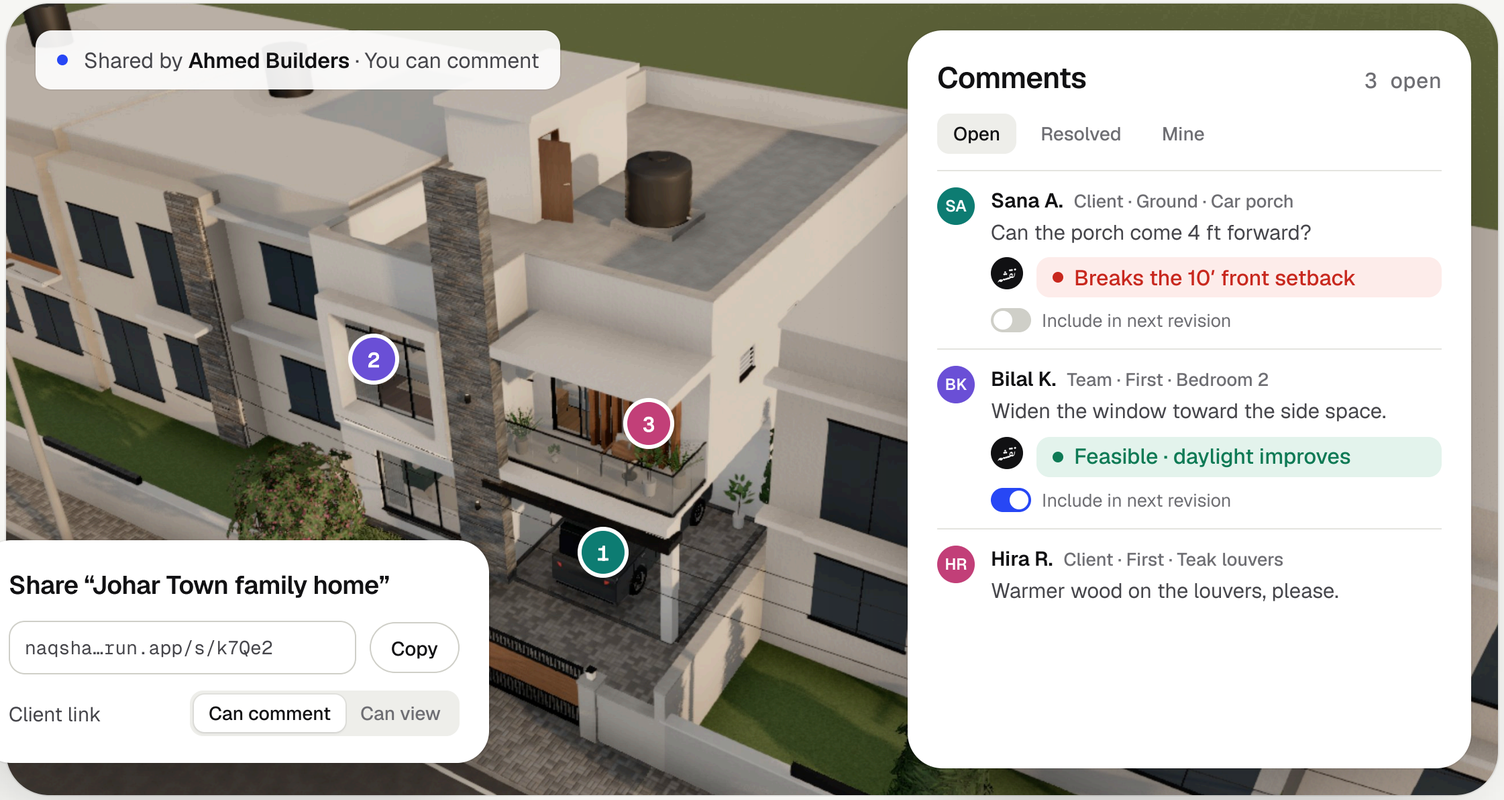
Every comment has an author: a name, a role and a colour, with replies. Naqsha answers with a feasibility chip.

The owner decides what goes into the next revision. Team comments go in by default; client comments wait for the owner.

Accounts on Google Identity Platform. A guest's homes move into the account on sign-in.

The client's view of a shared home

○ In the 18 Oct build



For contrast, maket.ai offers a view-only link and nothing else (tested 16 Oct 2026).

Roles: Owner and Team edit and revise · Client comments and approves · Viewer looks

Free to imagine. Paid to build. Lahore first, then Punjab.

One engine for four audiences. A new authority is a rule file, not a rebuild.

Families	See, walk and agree on the house before paying for drawings, and catch rule breaks before a resubmission that repeats a ≈ Rs 38,000 scrutiny fee.
Architects and builders	Win the client in the first meeting with a walkable concept and an LDA pre-check.
Developers and schemes	A sample home for every plot size in a scheme, to sell plots with.
PCATP architects, as stamping partners	They review, stamp and submit a pre-checked plan. A 1:100 print set, the scale LDA asks for, is in the 18 Oct build.

How it grows across Punjab · rules are data



An authority switch is built in: DHA height and storey caps today.

Pricing idea, to test in pilots · no prices set yet

Free

The brief, the LDA envelope and four concepts.

free

Per home

Plan, 3D, walkthrough, revisions and a print set.

paid by the family

Pro seats

Client links, branded title blocks, many homes, and a referral to a stamping architect.

per seat, by the firm

For reference, maket.ai charges $20 a month (Homeowner) and $100 a month (Pro).

Pilot metrics to track: first-submission acceptance, days to a first drawing, revision rounds

Every family in Punjab walks through a legal home before paying for drawings

Now

the 18 Oct 2026 submission build

Livable-home rules and Argon's self-review	○ 18 Oct
Collaboration: roles, client links, replies	○ 18 Oct
Day and night concepts	○ 18 Oct
A per-floor room program	○ 18 Oct
Solar-ready roofs, a choice of water tank	○ 18 Oct
Print to PDF at 1:100	○ 18 Oct

Milestone

Every item verified on the live URL.

Next

depth and reach

- Drag-to-swap rooms with an instant LDA re-check
- Google sign-in
- A sandboxed iframe for generated code
- Run events in Firestore, resumable anywhere
- Rule packs for RDA, FDA and GDA, then DHA and Bahria Town

Milestone

Rule packs live for three more authorities.

Later

ideas, not yet specified

- Pilots with LDA-enlisted PCATP architects
- Submission-ready sets for LDA's online system
- Structural and services drafts
- An Urdu interface
- Cost estimates

Milestone

A first pilot with practising architects.

Status: ● Live on the service today · ○ 18 Oct in the submission build, not yet verified live

Naqsha نقشہ

Walk through your home before the first brick.

Naqsha is live. Open a sample home in a minute, or design your own in about seven.

It said no to the porch, and told us which rule.

Live	naqsha-242926648187.us-central1.run.app
Code	github.com/abdm693/hack2skill-hackathon
Demo	[video URL, under 3 min]

What we'd build next on Google Cloud

- Run events in Firestore with Cloud Tasks, so any instance can resume a design
- Rule packs for RDA, FDA and GDA, then DHA and Bahria Town
- A pilot with LDA-enlisted PCATP architects and Lahore builders

photo photo photo

[Team name]

[Name, role] · [Name, role] · [Name, role]

